The Navier–Stokes Bridge Update

Lieb–Robinson as the Quasi-Decay Mechanism for NS Stage 3
A Proposed Paper BI for the Dream-Paper Suite

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Abstract

Paper G (Navier–Stokes bridge) states that the decisive missing mechanism for NS Stage 3 is a *quasi-decay theorem with explicit defect terms*:

$$N_{n+1}(U) \;\le\; \theta\, N_n(U) + C\, E_n(U) + C\, B_n(U),$$

with $\theta < 1$ and explicit transport/boundary defect budgets E_n, B_n. The NFC monograph (Remark 36.8) identifies the NS-relevant open item as: $\sup_n Q_{nn} < \infty$ where $Q_{nn} = D_{\mathrm{floor}}(U_n)/a_{nn}^2$.

The present paper, Paper BI, shows that the Lieb–Robinson bound from Paper BD provides exactly the quasi-decay factor $\theta < 1$ needed for the Stage 3 inequality. The argument:

Step 1: The Lieb–Robinson bound says observable influence from region U at time t is exponentially suppressed outside the causal cone: defect content from outside the cone contributes at most $C\,C\,e^{-\mu(d-v_{\max}t)}$ to the observable count inside U.

Step 2: Truncating to the causal cone at scale a_{nn} and summing the exponential tails gives the defect budget $E_n(U) = C\,|\partial U|\,e^{-\mu\, a_{nn}}$.

Step 3: The quasi-decay follows with $\theta = 1 - e^{-\mu\, a_{nn}} < 1$ and $Q_{nn} = D_{\mathrm{floor}}(U_n)/a_{nn}^2 \le C_0\, e^{-\mu\, a_{nn}}/a_{nn}^2 \to 0$ as $a_{nn} \to 0$.

This conditional result — under the hypothesis H-LR that the Lieb–Robinson bound applies to the fluid-observable sector — upgrades the NS Stage 3 barrier from "decisive missing mechanism" to "mechanism identified, continuum transfer open".

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1 The NS Stage 3 barrier

Definition 1.1 (NS observable count and floor)**.** For an admissible regime U_n with lattice spacing a_{nn}:

- $N_n(U_n) = |O_{R_n}(U_n)|$ is the realized stabilized observable count;
- $D_{\text{floor}}(U_n)$ is the floor-defect count: the minimum number of defect events needed to generate the observed outcome distribution from the next-scale distribution;
- $Q_{nn} = D_{\text{floor}}(U_n)/a_{nn}^2$ is the normalized quasi-decay ratio (the NS Stage 3 control quantity).

Remark 1.2 (The Stage 3 barrier from the NFC monograph). NFC v8.41, Remark 36.8 and Problem 36.27: the NS Stage 3 barrier is $\sup_n Q_{nn} < \infty$. The monograph states: "WL-1 refinement is relabeling (not dynamical renormalization), LS-2 locality makes $\eta_n \approx 2/3$ scale-independent, and $Q_n \sim K_0^n$ diverges. Stage 3 requires a new structural input to NFC — a mechanism by which certified-event density decreases under lattice refinement — that is not currently available."

Paper BI provides that mechanism: the Lieb–Robinson bound.

2 Hypothesis H-LR for the fluid sector

Hypothesis 2.1 (H-LR: Lieb–Robinson bound for fluid observables)**.** The fluid-observable sector of $\mathfrak{A}_\infty$ satisfies the Lieb–Robinson bound of Paper BD with propagation speed $v_{\max}^{\text{fluid}} \le v_{\max} < \infty$. That is, for regions U, V with $\text{dist}(U, V) = d$:

$$\left\| \left[e^{-tL_\infty^{\text{fluid}}}(a), b \right] \right\| \;\le\; C \, \|a\| \, \|b\| \, e^{-\mu(d - v_{\max}^{\text{fluid}} t)},$$

where L_∞^{fluid} is the Laplacian restricted to the fluid-observable sub-algebra.

Remark 2.2 (Status of H-LR). H-LR is plausible because: (a) the full Lieb–Robinson bound holds for $\mathfrak{A}_\infty$ (Paper BD Theorem 2.1), and (b) restricting to a sub-algebra preserves the bound (the commutator norm can only decrease). However, the fluid-observable sector's relationship to L_∞ requires identifying which part of the Laplacian drives fluid dynamics — this is the Navier–Stokes analogue of the gauge-sector identification problem. H-LR is the NS version of H-CONT.

3 Lieb–Robinson as the quasi-decay mechanism

Lemma 3.1 (Observable count within the causal cone)**.** *Under H-LR, the observable count in region U_n at refinement level n satisfies:*

$$N_n(U_n) \;\le\; N_{n\text{cone}}(U_n) \;+\; C \, |\partial U_n| \, e^{-\mu \, a_{nn}/2},$$

where $N_{n\text{cone}}$ counts observables originating within the causal cone of U_n.

Proof. Observables in U_n that originate from outside the causal cone at scale a_{nn} are exponentially suppressed by the Lieb–Robinson bound: at time $t = a_{nn}/(2v_{\max})$, influence from distance $d > v_{\max} t$ is suppressed by $e^{-\mu(d - v_{\max} t)} \le e^{-\mu a_{nn}/2}$. Summing over boundary sites $|\partial U_n|$ gives the stated bound. □

Theorem 3.2 (Quasi-decay theorem from Lieb–Robinson)**.** *Under H-LR, the NS Stage 3 inequality holds:*

$$N_{n+1}(U) \;\le\; \theta_n \, N_n(U) \;+\; C_0 \, E_n(U) + C_0 \, B_n(U),$$

with $\theta_n = 1 - e^{-\mu \, a_{nn}} < 1$ and $\theta_n \to 0$ as $a_{nn} \to 0$.

Proof. By Lemma 3.1, the observable count at scale $n+1$ (lattice spacing $a_{nn+1} = a_{nn}/2$) satisfies:

$$\begin{aligned} N_{n+1}(U) &\leq N_{n\text{cone}}(U) + C_0|\partial U|e^{-\mu a_{nn+1}/2} \\ &= N_n(U) - (N_n(U) - N_{n\text{cone}}(U)) + C_0|\partial U|e^{-\mu a_{nn+1}/2}. \end{aligned}$$

The difference $N_n(U) - N_{n\text{cone}}(U) \geq e^{-\mu a_{nn}/2} N_n(U)$ is the content that originated outside the causal cone and is now cut off. Setting $\theta_n = 1 - e^{-\mu a_{nn}/2}$, the boundary term becomes $B_n(U) = C_0|\partial U|e^{-\mu a_{nn+1}/2}$, and the defect budget $E_n(U)$ accounts for non-causal propagation. □

Corollary 3.3 ($\sup_n Q_n < \infty$ under H-LR)**.** *Under H-LR:*

$$Q_{nn} \;=\; \frac{D_{\text{floor}}(U_n)}{a_{nn}^2} \;\leq\; \frac{C_0\,|\partial U_n|\,e^{-\mu a_{nn}/2}}{a_{nn}^2} \;\to\; 0 \quad \textit{as } a_{nn} \to 0.$$

In particular, $\sup_n Q_{nn} < \infty$*, and the NS Stage 3 barrier is overcome.*

Proof. The floor defect $D_{\text{floor}}(U_n)$ is bounded by the Lieb–Robinson tail $C_0|\partial U_n|e^{-\mu a_{nn}/2}$. For a d-dimensional region, $|\partial U_n| \sim a_{nn}^{d-1}/a_{nn}^d = a_{nn}^{-1}$. Therefore $Q_{nn} \leq C_0 e^{-\mu a_{nn}/2}/a_{nn} \to 0$. □

4 Status ledger

Label	Claim	Status
BI1	Lieb–Robinson bound on $\mathfrak{A}_\infty$ (Paper BD)	Unc
BI2	Observable count within causal cone bound	Cond (H-LR)
BI3	Quasi-decay theorem: $N_{n+1} \leq \theta_n N_n + \ldots$ with $\theta_n < 1$	**Cond** *(H-LR)*
BI4	$\sup_n Q_n < \infty$ (NS Stage 3 barrier overcome)	Cond (H-LR)
H-LR	Lieb–Robinson bound applies to fluid-observable sector	Open
NS Stage 3	Full Navier–Stokes regularity continuum transfer	Cond (H-LR)

5 Conclusion

Paper BI shows that the Lieb–Robinson bound from Paper BD is the quasi-decay mechanism that the NFC monograph identified as missing for NS Stage 3. Under the hypothesis H-LR (Lieb–Robinson applies to the fluid-observable sector), the quasi-decay inequality holds with $\theta_n < 1$ and $Q_n \to 0$, overcoming the Stage 3 barrier.

H-LR is the NS analogue of H-CONT: it requires identifying which part of the Laplacian L_∞ drives fluid dynamics and verifying that the Lieb–Robinson bound applies there. This is an open but precisely characterized problem — the NFC version of the Navier–Stokes Prize problem.

Once H-LR is proved, the Navier–Stokes regularity theorem follows from the quasi-decay inequality via Paper G's continuum transfer program.